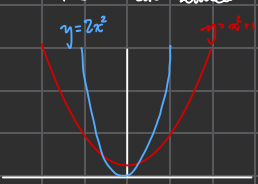


10.1: Finding Areas:

Suppose 2 curves: $f(x)$ & $g(x)$. Subtracting these from each other gives the difference in height function between the curves.

Hence integrating this function (with potential bounds of intersection) gives the area of the difference of height function, or the areas between the curve.



$$\Rightarrow \Delta h = y = 1-x^2 = \overbrace{(1+x)(1-x)}^{\Delta h = 0}$$

$$\Rightarrow \int_{-1}^{1}(1-x^2)\,dx = \Delta A.$$

Integration acts as a ways to calculate Riemannian sums which are infinitesimally thin (δx). This is multiplied by the height ($f(x)$).

Hence: $\delta A = f(x)\,\delta x$

$$\Rightarrow A = \sum_i \delta A = \sum_{x=a}^{b} f(x)\,\delta x$$

For an accurate integral:

$$\Rightarrow A = \lim_{\delta x \to 0} \sum_{x=a}^{b} f(x)\,\delta x = \int_a^b f(x)\,dx$$

To get Area between the curve & $x=0$, rearrange in terms of y, and integrate normally ($\int x\,dy$)

10.2: Integration by Substitution:

Groups terms of variables into another variable, and integrate with respect to the new variable (hence dx has to be found). If it is a definite integral, adjust the bounds in terms of the new variable ($u = 3x+1 \Rightarrow x=1 \Leftrightarrow u=2$)
Can also be done by recognition, knowing you will divide through by the derivative of 'x'.

$$\Rightarrow \int (2x+4)^4\,dx \quad \left[\begin{array}{l}\text{doing diff. these would go down } (5\to 4)\\ \text{Hence } \frac{1}{(4+1)2} = \frac{1}{10}\end{array}\right] = \frac{1}{10}(2x+4)^5 \ (+C)$$

Can differentiate to Check!

Remember!

$$\int g'(f(x))\,dx \neq \frac{1}{f'(x)}\,g(f(x)) + C$$

only works for linear $f(x)$

10.3: Integrating other functions:

$$\int f'(x)\sin(f(x))\,dx = -\cos(f(x)) + C$$

$$\int f'(x)\cos(f(x))\,dx = \sin(f(x)) + C$$

$$\int f'(x)\sec^2(f(x))\,dx = \tan(f(x)) + C$$

$$\int f'(x)\tan(f(x))\,dx = -\ln\left|\cos(f(x))\right| + C$$

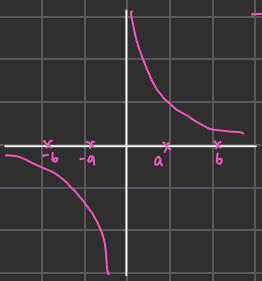
$$\int f'(x)e^{f(x)}\,dx = e^{f(x)} + C$$

All trivial & easily provable.

10.4: Integrating with the Natural Logarithm:

$$\frac{d}{dx}\left(\ln(f(x))\right) = \frac{f'(x)}{f(x)} \Rightarrow \int \frac{f'(x)}{f(x)}\,dx = \ln(f(x)) + C$$

However...



$$-\int_{-b}^{-a} \frac{1}{x}\,dx, \quad u = -x$$

$$\Rightarrow du = -dx$$

$$\Rightarrow \int_{b}^{a} \frac{1}{-u}\,du = -\int_{b}^{a} \frac{1}{u}\,du = \int_{a}^{b} \frac{1}{u}\,du = \ln b - \ln a = \int_{a}^{b} \frac{1}{x}\,dx$$

Hence: $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$

U-Sub & Recognition can be used.

10.5: Further Integration by Substitution:

Some substitutions can work, even if not by recognition

10.6: Integration by parts:

The product rule states:

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}, \text{ where } u = u(x),\ v = v(x)$$

$$\Rightarrow d(uv) = u\,dv + v\,du$$

$$\Rightarrow \int d(uv) = \int u\,dv + \int v\,du$$

$$\Rightarrow \int u\,dv = uv - \int v\,du$$

Integrals with a product within can be solved this way.
Hence u has to be found & differentiated.
$\frac{dv}{dx}$ must be found & integrated (w.r.t x)

To find u, follow:

Logs
Inverse Trig
Algebra
Trig
Exponentials

$\frac{dv}{dx}$ takes the other left over term

For definite integration:

$$\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du$$

Worked Example:

$$\int xe^x\,dx$$

$u = x$ $\quad \frac{dv}{dx} = e^x$

$\Rightarrow du = dx$ $\quad \Rightarrow v = e^x$

$\left(\frac{dy}{dx} = 1\right)$

$$\Rightarrow xe^x - \int e^x dx = xe^x - e^x + C$$

Integrating $\ln x$

$$\int \ln x\,dx = \int 1 \cdot \ln x\,dx$$

$u = \ln x$ $\quad \frac{dv}{dx} = 1$

$\Rightarrow du = \frac{1}{x}dx$ $\quad \Rightarrow v = x$

$$\Rightarrow \int \ln x\,dx = x\ln x - \int dx = x\ln x - x + C$$